L.J Institute of Engineering and Technology

Data Structures using JAVA (SEM-II EVEN_2025-CE RELATED BRANCHES)

Note :

This practice book is only for reference purpose . L.J.U Test question paper may not be completely set from this practice book.

Date : 7 JUL 2025

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
1	9	The pre-order traversal of a Binary search tree is 15,10,12,11,20,18,16,19. Which one of the following is the post order traversal of the tree?	B	1	10,11,12,15,16,18,19,20	11,12,10,16,19,18,20,15	20,19,18,16,15,12,11,10	19,16,18,20,11,12,10,15
2	9	In delete operation of BST, we need inorder successor (or predecessor) of a node when the node to be deleted has both left and right child as non-empty. Which of the following is true about inorder successor needed in delete operation?	B	1	Inorder Successor is always a leaf node	Inorder successor is always either a leaf node or a node with empty left child	Inorder successor may be an ancestor of the node	Inorder successor is always either a leaf node or a node with empty right child
3	9	Which of the following traversals is sufficient to construct BST from given traversals 1) Inorder 2) Preorder 3) Postorder	B	1	Any one of the given three traversals is sufficient	Either 2 or 3 is sufficient	2 and 3	1 and 3
4	9	A binary search tree is generated by inserting in following order: 50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24 The number of the node in the left sub-tree and right sub-tree of the root, respectively, is	B	1	(4, 7)	(7, 4)	(8, 3)	(3, 8)
5	9	Suppose the numbers 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are inserted in that order into an initially empty binary search tree. The binary search tree uses the usual ordering on natural numbers. What is the in-order traversal sequence of the resultant tree ?	C	1	7 5 1 0 3 2 4 6 8 9	0 2 4 3 1 6 5 9 8 7	0 1 2 3 4 5 6 7 8 9	9 8 6 4 2 3 0 1 5 7
6	9	While inserting the elements 71, 65, 84, 69, 67, 83 in an empty binary search tree (BST) in the sequence shown, the element with highest value of depth is	B	1	65	67	69	83
7	9	How many distinct binary search trees can be created out of 5 distinct keys?	D	1	14	26	34	42
8	9	How many distinct binary search trees can be created out of 4 distinct keys?	B	1	4	14	24	42
9	9	Which of the following traversal outputs the data in sorted order in a BST?	B	1	PreOrder	Inorder	Postorder	LevelOrder
10	9	Which traversal of Binary search tree gives an ascending order	D	1	Preorder	Post order	A & B Both	Inorder
11	9	Consider a binary search tree T with distinct keys. The right sub tree of a node x in T is empty and the node x has an inorder successor y, then minimum no. of ancestors node x is having is_____? (every node is considered to be its own ancestor)	D	1	4	5	6	2

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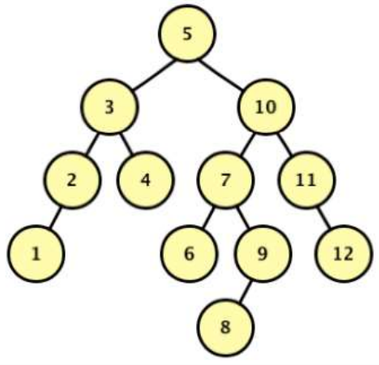
12	9	The number 1,2.....n are inserted in a binary search tree in some order, in the resulting tree, the right subtree of the root contains P nodes. the first number to be inserted in the tree must be	C	1	P	P+1	N-P	P=1
13	9	While inserting the elements 72, 65, 86, 69, 66, 81,23,7,89 in an empty binary search tree (BST) in the sequence shown, the elements in the left subtree and the elements in the right subtree are_____	B	1	3 and 5	5 and 3	5 and 4	4 and 5
14	9	A binary search tree is constructed by inserting the following numbers in order : 60, 25, 72, 15, 30, 68, 101, 13, 18, 47, 70, 34 The number of nodes in the left sub tree is :	A	1	7	2	3	4
15	9	Let T be a binary search tree with 63 nodes. The minimum and maximum possible heights of T are:	C	1	3 and 62	4 and 62	5 and 62	6 and 62
16	9	Let T be a binary search tree with 15 nodes. The minimum and maximum possible heights of T are:	B	1	4 and 15 respectively	3 and 14 respectively	4 and 14 respectively	None of the above
17	9	Suppose numbers from 1 to 1000 are saved in a binary search tree and we want to find 363. Which of the following sequences cannot be the order of elements while reaching the searched value?	A	1	925,202,911,240,912,245,363	924,220,911,244,898,258,362,363	2,252,401,398,330,344,397,363	2,399,387,219,266,382,381,278,363
18	9	Suppose that we have numbers between 1 and 100 in a binary search tree and want to search for the number 55. Which of the following sequences CANNOT be the sequence of nodes examined?	C	1	{10, 75, 64, 43, 60, 57, 55}	{90, 12, 68, 34, 62, 45, 55}	{9, 85, 47, 68, 43, 57, 55}	{79, 14, 72, 56, 16, 53, 55}
19	9	The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)?	B	1	2	3	4	6
20	9	If we delete node 10 from given below BST with its in order successor node then what will be the height of the resultant BST and which element has height 4 respectively?  	B	1	4,8	4,5	4,6	3,6

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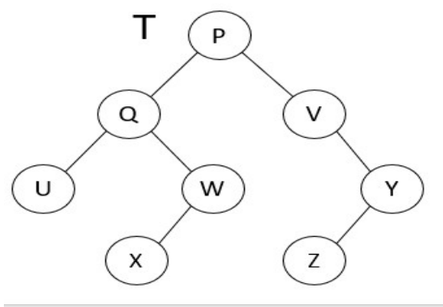
21	9	Consider the following binary search tree T given below: Which node contains the fourth smallest element in T? (Here each variable represents name of node and not the value node contains) 	D	1	U	Q	X	W
22	9	Create a Binary Search Tree for the following data and do in-order, Preorder and Post-order traversal of the tree. 40, 60, 15, 4, 30, 70, 65, 10, 95, 25, 34		5				
23	9	Create a BST from the given traversal. Postorder: 1, 11, 12, 10, 14, 18, 21, 17, 15, 13		5				
24	9	Create a binary search tree for the following data : 50,25,75,22,40,60,80,90,15,30		5				
25	9	Given the following traversal create a binary search tree from that. Also give the postorder traversal for the same. preorder = {7,4,3,1,2,10,8,11}		5				
26	9	Write down a Boolean return type method **"findElement(node root, int value)"** that returns true if a particular value is available in in Binary Search Tree else it return false.		2				
27	9	Write down a function to insert element in Binary Search Tree.		3				
28	9	Construct a binary search tree for the following and perform inorder and postorder traversals: 5 9 4 8 2 1 3 7 6		5				
29	9	The postorder traversal of a binary search tree is 25,33,30,35,42,48,40,60,58,50. The inorder traversal of the same tree is 25,30,33,35,40,42,48,50,58,60. What is the length of the longest path from one leaf to another leaf. (Note: Length of longest path means total number of nodes present in that path). Note: Show the Tree as well		1				
30	9	Construct binary search tree for the following data and perform in-order and post-order traversals: - 50, 40, 80, 20, 0, 30, 10, 90, 60, 70		5				
31	9	Insertion sequence of names is Norma, Roger,John,Bill,Leo,Paul, Ken and Maurice. Create a BST. (i) Insert Kirk. Show the BST. (ii)Delete John. Show the BST		5				
32	9	Create a binary search tree for inserting the following data. 50, 45, 100, 25, 49, 120, 105, 46, 90, 95 and Delete 49		5				
33	9	Create a BST from the given traversal. Preorder: 50, 25, 24, 33, 48, 60, 55, 79, 99		5				

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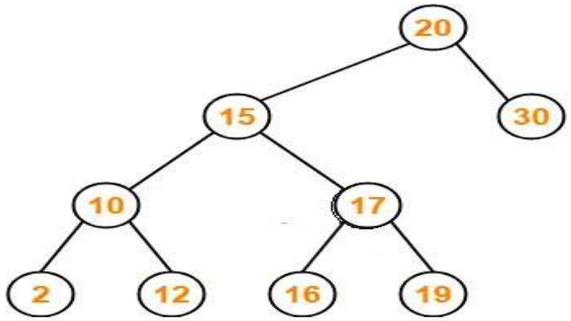
34	9	Construct a binary search tree for the following sequence. also do the Inorder and Postorder traversal for the same. 45,56,39,12,34,78,54,67,10,32,89,81		2				
35	9	Create a Binary Search Tree for the following data and do Inorder, Preorder and Postorder traversal of the tree.40, 65,25, 55, 10,70,30,50,15,80,75		5				
36	9	Construct binary search tree for the following data: 10,3,15,22,6,45,65,23,78,34,5 Answer the following questions - A) Number of leaves in the given BST B) Height of the BST obtained from above elements C) Number of nodes in Left and Right subtree of root in the BST respectively D) Draw the final BST after deleting 15, 5, 45 considering the right subtree is		5				
37	9	Construct a binary search tree from the traversals given below: Inorder: 1, 10, 11, 12, 13, 14, 15, 17, 18, 21 Postorder: 1, 11, 12, 10, 14, 18, 21, 17, 15, 13		5				
38	9	Construct BST from Preorder traversal: 25,15,10,4,12,22,18,24,50,35,31,44,70,66,90. Also write down post order traversal		2				
39	9	Consider the following BST where delete node with value = 15 from its Predecessor node.  		2				
40	9	Pre-order of BST is given as follow, develop a BST, also perform deletion of H, M, and O in the given sequence. Pre-order = H, G, L, K, O, M, Q		2				
41	9	Create a binary search tree for inserting the following data : 50, 45, 100, 25, 49, 120, 105, 46, 90, 95 Explain deleting node 45 in the resultant binary search tree.		5				
42	9	What is the maximum height of any height balanced BST tree with 20 nodes?	C	1	3	4	5	6
43	9	Which of the following is true about AVL?	C	1	AVL trees are self-balancing binary search trees.	In AVL trees, balancing factor of each node is either 0 or 1 or -1.	Both	None of the above
44	9	The number of rotations required to insert a sequence of elements 28, 73, 89, 75, 74, 13 into an empty AVL tree is?	C	1	1	5	2	3
45	9	How many rotations are required during construction of an AVL tree if the following elements are added in order given? 35, 50, 40, 25, 30, 60, 78, 20, 28	B	1	1	3	4	5

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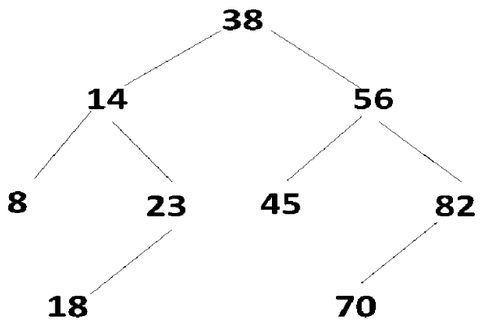
46	9	Consider the following AVL tree. If we insert 20 in the below AVL tree then which node will become imbalance first from the bottom and which rotation will be performed for the height balanced AVL tree? 	B	1	38, LL	23, LR	23, RL	38, LR
47	9	Consider following statement: S1: Rotation operation in AVL always preserves the Inorder ordering. S2: Rotation operation in AVL always preserves the pre order ordering. S3: Rotation operation in AVL always preserves the post order ordering.True Statements ?	A	1	S1	S1,S2	S1,S2,S3	None of these
48	9	What is the maximum height of any AVL-tree with 7 nodes? Assume that the height of a tree with a single node is 0.	B	1	2	3	4	5
49	9	In what order we should insert the following elements into an empty AVL tree so that we don't have to perform any rotation on it. 1, 2, 3, 4, 5, 6, 7	B	1	4, 2, 1, 6, 3, 5, 7	4, 2, 6, 1, 3, 5, 7	6, 4, 5, 7, 1, 2, 3	4, 5, 3, 2, 1, 6, 7
50	9	The number of rotations required to insert a sequence of elements 9, 6, 5, 8, 7, 10 into an empty AVL tree is?	D	1	0	1	2	3
51	9	Given an empty AVL tree, how would you construct AVL tree when a set of numbers are given without performing any rotations?	B	1	just build the tree with the given input	find the median of the set of elements given, make it as root and construct the tree	use trial and error	use dynamic programming to build the tree
52	9	Consider the below left-left rotation pseudo code where the node contains value pointers to left, right child nodes and a height value and Height() function returns height value stored at a particular node. avltree leftrotation(avltreenode x): avltreenode w = x.left x.left=w.right w.right=x x.height=max(Height(x.left),Height(x.right))+1 w.height=max(missing)+1 return w	A	1	Height(w.left), x.height	Height(w.right), x.height	Height(w.left), x	Height(w.left)
53	9	To restore the AVL property after inserting a element, we start at the insertion point and move towards root of that tree. is this statement true? a) true b) false	A	1	true	false	Both depending on conditions	None of the above

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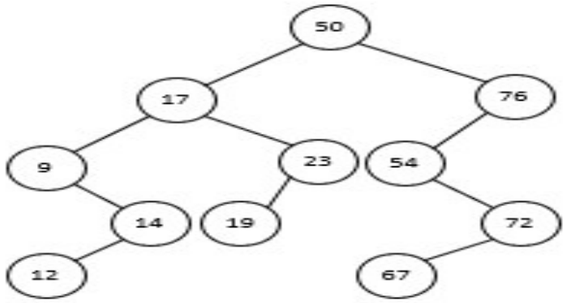
54	9	Given tree is balanced binary search tree or not? Find out Balanced factor of each node. Which is the unbalanced node in the given tree. 		2				
55	9	Obtain an AVL tree by inserting one integer at a time in the following sequence. 150, 155, 160, 115, 110, 140, 120, 145, 130, 147, 170, 180. Show all the steps. Also show the total number of rotations needed of type 1) L-L 2) R-R 3) L-R and 4) R-L		5				
56	9	1. Construct an AVL Tree with the following elements and mention the actions taken along with all rotations (L, R, LR, RL) clearly drawn. H, I, J, B, A, E, C, F, D, G, K, L 2. Delete F, E in order from the above AVL tree.		4				
57	9	1.Create a BST with the following data -> 10, 1, 3, 5, 15, 12, 16 2. Delete the node 10 from the tree such that there should be no changes in the left subtree of 10 3. Insert a new node 7 4. Find Preorder of Final BST.		3				
58	9	Construct a height balanced binary tree (AVL tree) for the following data and show all steps with rotations 42,06,54,62,88,50,22,32,12,33		5				
59	9	Obtain the AVL tree by inserting one integer at a time in the following sequence. 45, 50, 55, 10, 5, 35, 15, 40, 25, 42, 65, 75. Show all the steps.		5				
60	9	Construct AVL tree for following data 10,20,30,40,50,60,70,80		5				
61	9	Construct the AVL search tree by inserting the following elements in the order of their occurrence. 64, 1, 44, 26, 13, 110, 98, 85		5				
62	9	Create an AVL tree for following sequence of numbers. 14,17,11,7,53,4,13,12,8,60,19,16,20		2				
63	9	Obtain the AVL tree by inserting one alphabet at a time in the following sequence. A, Z, B, Y, C, X, D, U, E		5				

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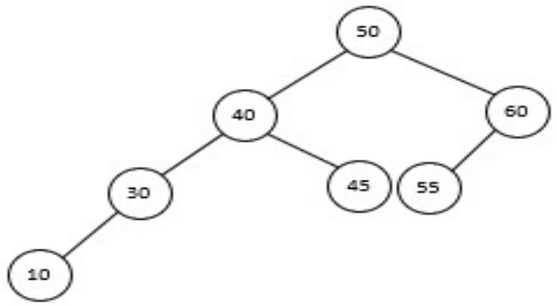
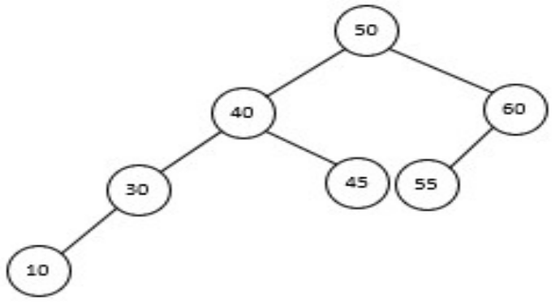
64	9	Delete Node 40 from the AVL tree shown in the following image. 		5				
65	9	Insertion sequence of names is Kavery, Bina, Gita, Seema, Hina, Zarana, Jay, Kavit, Kavya, Mayuri, Anil, Bijal, Mina, Sheela, , Anuj. Construct BST and show the behavior of creating a lexically ordered binary tree. 1. Write down Preorder and Post order traversal of BST 2. Show Balance factor on each node. 3. Delete Seema by Inorder predecessor and show BST. 4. Insert Kapil and Show BST		2				
66	9	Obtain an AVL tree by inserting one integer at a time in the following sequence. 150, 155, 160, 115, 110, 140, 120, 145, 130, 147, 170, 180. Show all the steps.		5				
67	9	Deleting 55 from the AVL Tree and generate again final AVL tree. 		2				
68	9	Construct AVL Tree for: 21,26,30,9,4,14,28,18,15,10,2,3,7		3				
69	9	Insert following elements into an AVL tree, after insertion of all element delete 35 from the AVL tree. elements - 32,8,3,71,35,24,27,37.		2				
70	9	Construct AVL tree for the following numbers 14, 8, 12, 46, 23, 5, 77, 88, 20.		2				
71	9	Insert the following sequence of elements into an AVL Tree Starting with an empty Tree 10, 20, 15, 25, 30, 16, 18, 19.		2				

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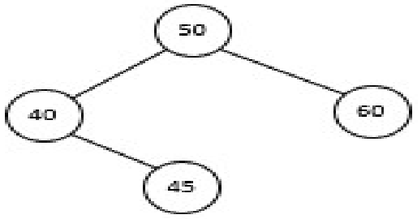
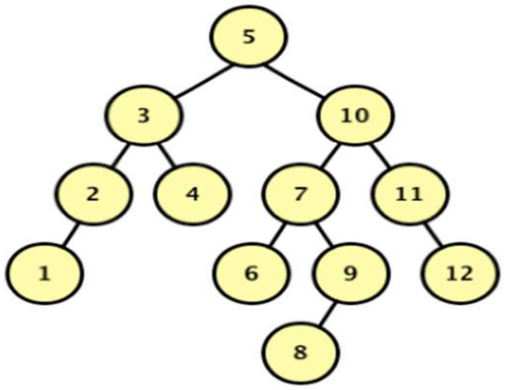
72	9	Delete the node 60 from the AVL tree shown in the following image. 		2				
73	9	Delete node 5,3 from given AVL tree. 		2				
74	9	24 / \ 8 35 / \ / 4 18 28 / \ 12 21 What will be the root of tree after inserting 10 and then again after deleting 28.		3				
75	9	Consider the following elements inserted into an empty AVL tree in the following order 25, 10, 15, 17, 30, 35, 40, 21, 28 If [Ld] be the sum of elements on left side of root and (Rd) be the sum of elements on right side of root, then the value of [(Rd) – (Ld) + Root] is _____.		3				
76	9	Find the maximum possible height of AVL tree with 87 nodes.		3				
77	9	2-3 tree is a specific form of _____	A	1	B tree	B+ tree	AVL tree	Heap
78	9	Elements 1,2,3,4,5,6,7,8,9,10 are inserted into 2-3 Trees in the same order. What is the maximum number of splits occurring after inserting all these keys into the tree?	C	1	1	2	5	4

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79	9	Which of the following is false?	A	1	2-3 tree requires less storage than the BST	lookup in 2-3 tree is more efficient than in BST	2-3 tree is shallower than BST	2-3 tree is a balanced tree
80	9	Out of these, which one is NOT true about a 2-3 tree?	D	1	it is perfectly balanced	the leaves are always at the same level	it refers to a B-tree of the order 3	postorder traversal would yield the elements in a sorted order
81	9	Create 2-3 Tree for the following sequence: 50, 100, 150, 200		5				
82	9	Construct 2-3 tree for the following data 50, 25, 1, 45, 7, 89, 44, 56, 23, 77		5				
83	9	Construct 2-3 tree for the following data 12, 50, 85, 6, 10, 37, 100, 120, 25, 70		5				
84	9	Create 2-3 tree for following sequence and count number of splitting operations required for resultant 2-3 tree. 10,9,8,7,6,5,4,3,2,1,11,12,13,14,15		3				
85	9	Which of the following statements are false about B tree of order m?	C	1	a node is full if it has m -1 keys	the internal node except the root node has atleast m/2 child nodes	leaf nodes are not at same level	all of above
86	9	Insert the following letters in B tree of order 5 C,N,G,A,H,E,K,Q,M,F,W,L,T,Z,D,P,R,X,Y,S,B		2				
87	9	Insert the following letters into an empty B-tree of order 5. C N G A H E K Q M F W L T Z D P R X Y S		2				
88	9	Construct B tree of order 5 for the following data: 1,12,8,2,25,6,14,28,17,7,52,16,48,68,3,26,29,53,55,45,67.		3				
89	9	Insert the following elements in a B-tree. A, G, F, B, K, D, H, M, J, E, S, I. R, X, C, L, N, T, U, P given order is 5.		3				
90	9	Construct B tree of order 5 for the following data 1,7,6,2,11,5,10,13,12,20,16,24,3,4,18,19,14,25.		5				
91	9	Insert the following elements in a B-Tree with order 3. a, g, f, b, k, c, h		2				
92	9	Insert the following elements in a B-Tree with order 4. a, g, f, b, k, c, h, n, j		5				
93	9	Construct B tree of order 5 for following data 1,6,7,2,11,5,10,13,12,20,16,24,3,4,18,19,14,25		2				
94	9	Construct a B-tree of order 3 by inserting numbers from 1 to 10		3				
95	9	A B+ tree can contain a maximum of 7 pointers in a node. What is the minimum number of keys in leaves?	B	1	6	3	4	7
96	9	Which of the following is false?	A	1	A B+ -tree grows downwards	A B+ -tree is balanced	In a B+ -tree, the sibling pointers allow sequential searching	B+ -tree is shallower than B-tree
97	9	Which of the following is true?	C	1	B + tree allows only the rapid random access	B + tree allows only the rapid sequential access	B + tree allows rapid random access as well as rapid sequential access	B + tree allows rapid random access and slower sequential access
98	9	In a B+ tree, both the internal nodes and the leaves have keys.	B	1	TRUE	FALSE	Sometimes	Both 1 and 3
99	9	Create B+ tree of order 5 using data given bellow 10,12,15,18,19,20,21,22,23,30,31,33,45,47,48,50,52.		3				

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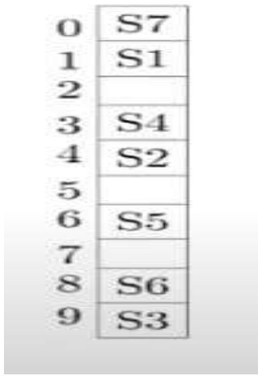
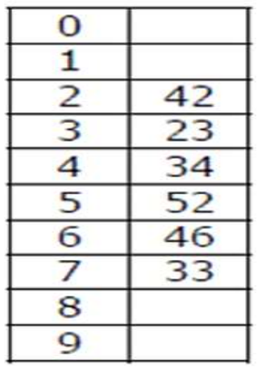
100	9	Insert the following key values 6, 16, 26, 36, 46 on a B+ tree with order = 3.		2				
101	10	What data organization method is used in hash tables?	C	1	Stack	Tree	Linked list	Queue
102	10	Which of the following techniques offer better cache performance?	B	1	Quadratic probing	Linear probing	Double hashing	Rehashing
103	10	What is the hash function used in the division method?	B	1	h(k) = k/m	h(k) = k mod m	h(k) = m/k	h(k) = m mod k
104	10	Let us consider a list of numbers (34, 16, 2, 93, 80, 77, 51) and has table size is 10. What is the order of elements (from index 0 to size-1) in the hash table?	C	1	null, null, 77, 16, null, 34, 93, 2, 51, 80	77, 16, 34, 93, 2, 51, 80	80, 51, 2, 93, 34, null, 16, 77, null, null	80, 51, 2, 93, 34, 16, 77
105	10	_____ is a collision-resolution scheme that searches the hash table sequentially, starting from the original location specified by the hash function, for an unoccupied location	A	1	Linear probing	Quadratic probing	Double hashing	Separate chaining
106	10	_____ is a collision-resolution scheme that uses an array of linked lists as a hash table.	D	1	Linear probing	Double hashing	Quadratic probing	Chaining
107	10	In the context of hashing, what is meant by a collision?	C	1	The hash function returns a value index greater than the table size.	The insertion algorithm cannot find an empty slot in the table.	Two equal objects hash to the same slot.	Two unequal objects hash to the same slot
108	10	Let us consider a list of numbers (34, 16, 2, 93, 80, 77, 51) and has table size is 8. What is the order of elements(from index 0 to size-1) in the hash table?	A	1	{16, 80, 34, 2, 51, 93, 77, NULL}	{51, 2, 93, 16, NULL, 77, 34, 80}	{ NULL, 51, 2, 93, 16, 77, 34, 80}	{ 77, 80, 34, 16, NULL, 93, 2, 51}
109	10	If the hash function is h(element) = element % 10 so which pairs of values the collision take place?	B	1	15 & 13	4 & 14	2 & 16	8 & 36
110	10	Consider a hash table of size seven, with starting index zero, and a hash function (3x + 4)mod7. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? Note that '_' denotes an empty location in the table.	B	1	8, _, _, _, _, _, 10	1, 8, 10, _, _, _, 3	1, _, _, _, _, _,3	1, 10, 8, _, _, _, 3
111	10	which one of the following hash functions on integers will distribute keys most uniformly over 10 buckets numbered 0 to 9 for ii ranging from 0 to 2020?	B	1	h(i)=i^2(mod10)	h(i)=i^3(mod10)	h(i)=(11*i^2)mod10	h(i)=(12*i)mod10
112	10	Using division method, in a given hash table of size 157, the key of value 172 be placed at position _____	C	1	19	72	15	17
113	10	Using division method, in a given hash table of size 156, the key of value 172 be placed at position_____.	C	1	15	11	16	2
114	10	An advantage of chained hash table (external hashing) over the open addressing scheme is	C	1	Worst case complexity of search operations is less	Space used is less	Deletion is easier	None of the above
115	10	Which hashing technique is free from clustering?	B	1	Linear Probing	Double hashing	Quadratic hashing	Rehashing
116	10	If several elements are competing for the same bucket in the hash table, what is it called?	C	1	Diffusion	Replication	Collision	Duplication
117	10	The integers given are to be inserted in a hash table with 5 locations using closed addressing/open hashing method to resolve collisions. Construct hash table and use division reminder method for hash function. 1,2,3,4,5,10,11,12,23,34,25,42,41,58,59,60		2				

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
118	10	Given the following input (4322, 1334, 1471, 9679, 1989, 6171, 6173, 4199) and the hash function x mod 10, which of the following statements are true? i. 9679, 1989, 4199 hash to the same value ii. 1471, 6171 hash to the same value iii. All elements hash to the same value iv. Each element hashes to a different value	C	1	i only	ii only	i and ii only	iii or iv
119	10	Consider a hash table with 9 slots. The hash function is h(k) = k mod 9. The collisions are resolved by chaining. The following 9 keys are inserted in the order: 5, 28, 19, 15, 20, 33, 12, 17, 10. The maximum, minimum, and average chain lengths in the hash table, respectively, are	A	1	3, 0, and 1	3, 3, and 3	4, 0, and 1	3, 0, and 2
120	10	The integers given are to be inserted in a hash table with 5 locations using chaining to resolve collisions. Construct hash table and use simplest hash function. 1,2,3,4,5,10,21,22,33,34,15,32,31,48,49,50.		7				
121	10	Build a chained hash table of 10 memory locations. Insert the keys 131, 3, 4, 21, 61, 24, 7, 97, 8, 9 in hash table using chaining. Use h(k) = k mod m. (m=10)		4				
122	10	Let us consider the insertions of elements 5,28,19,15,20,33,12,17,10 into a chained-hash table. Let us suppose the hash table has 9 slots and the hash function be h(k)= k mod 9.		4				
123	10	Consider inserting keys 78,58,22,21,85,88,90,65,82,10,28,52,64 into hash table of size m=10 using chaining.		2				
124	10	Consider a hash table with 11 slots and the collisions are resolved by linear probing. The following keys are inserted in the order: 15, 2, 1, 5, 20, 31, 12, 21, 17, 34. The hash function is h(k) = k mod 11. The hash table state will be,	D	1	{20,1,2,31,12,15,5,17,21,34,NULL}	{1,2,31,20,15,12,5,17,21,NULL, 34}	{NULL, 20,1,2,31,12,15,5,17,21,34}	{21,1,2,12,15,5,17,34,NULL,20, 31}
125	10	Consider a 13-element hash table for which f(key)=key mod 13 is used with integer keys. Assuming linear probing is used for collision resolution, at which location would the key 103 be inserted, if the keys 661, 182, 24 and 103 are inserted in that order?	B	1	0	1	11	12
126	10	A hash function h defined h(key)=key mod 7, with linear probing, is used to insert the keys 44, 45, 79, 55, 91, 18, 63 into a table indexed from 0 to 6. What will be the location of key 18 ?	C	1	3	4	5	6
127	10	Consider a hash table of size seven with starting index zero and a hash function (6x+3)mod4. Assuming the hash table is initially empty. Which of the following is the content of the table when the sequence 1,3,8,10,5, is inserted into the table using Linear probing? Here - denotes an empty location in the table.	D	1	1,3,8,10,5,_,_	3,8,1,_,10,5	_3,5,8,10,1,_	None of these
128	10	Consider hash table size is 5 and hash function is H(k) = k mod N. Given keys are 13, 25, 35, 45, 55, 65 and want to arranged in hash table with linear probing method then location of 65 will be in hash table is _____	D	1	1	2	3	This method is not suit for the given keys
129	10	A hash table contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is key % 10. If the values 43, 165, 62, 123, 142 are inserted in the table, in what location would the key value 142 be inserted.	D	1	1	2	5	6

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
130	10	A hash table with ten buckets with one slot per bucket is shown in the following figure. The symbols S1 to S7 initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is  	A	1	5	4	3	2
131	10	A hash table of length 10 uses open addressing with hash function h(k)=k mod 10, and linear probing. After inserting 6 values into an empty hash table, the table is as shown below.Which one of the following choices gives a possible order in which the key values could have been inserted in the table?  	C	1	46, 42, 34, 52, 23, 33	34, 42, 23, 52, 33, 46	46, 34, 42, 23, 52, 33	42, 46, 33, 23, 34, 52

<table>
<tr><th>Sr No</th><th>Unit Number</th><th>Question_Text</th><th>MCQ Answer</th><th>Marks</th><th>Option A</th><th>Option B</th><th>Option C</th><th>Option D</th></tr>
<tr><td>132</td><td>10</td><td>A hash table with eleven buckets with one slot per bucket is shown in the following figure. Some keys initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is ____?

<table><tr><th>location</th><th>keys</th></tr><tr><td>0</td><td>11</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>24</td></tr><tr><td>3</td><td>36</td></tr><tr><td>4</td><td></td></tr><tr><td>5</td><td></td></tr><tr><td>6</td><td>72</td></tr><tr><td>7</td><td>29</td></tr><tr><td>8</td><td>19</td></tr><tr><td>9</td><td>20</td></tr><tr><td>10</td><td>21</td></tr></table></td><td>B</td><td>1</td><td>7</td><td>10</td><td>1</td><td>5</td></tr>
<tr><td>133</td><td>10</td><td>The keys 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function h(k) = k mod 10 and linear probing. What is the resultant hash table?</td><td></td><td>7</td><td></td><td></td><td></td><td></td></tr>
<tr><td>134</td><td>10</td><td>Insert the key values 26,37,59,76,65,86 in to a hash-table of size m=11 using linear probing.consider the primary hash function is h'(k) = k mod m.</td><td></td><td>4</td><td></td><td></td><td></td><td></td></tr>
<tr><td>135</td><td>10</td><td>Consider inserting keys 78,58,22,21,85,88,90,65,82,10 into hash table of size m=12 using linear probing. Consider hash function h(k)=k%12.</td><td></td><td>2</td><td></td><td></td><td></td><td></td></tr>
<tr><td>136</td><td>10</td><td>Consider a hash table of size 11 with linear probing and a hash function h(K) = K mod N. The following keys are inserted in an empty hash table : 6, 17, 2, 63, 81, 40, 9, 28 Show the representation of hash table and give the list of empty indices after insertion.</td><td></td><td>2</td><td></td><td></td><td></td><td></td></tr>
<tr><td>137</td><td>10</td><td>Insert the key values 36,47,69,86,75,96 in to a hash-table of size m=11 using linear probing. Consider the primary hash function is h'(k) = k mod m.</td><td></td><td>2</td><td></td><td></td><td></td><td></td></tr>
<tr><td>138</td><td>10</td><td>Use quadratic probing method to insert following data elements : For a hash table, Table Size = 7
Keys = 22,30,50,35,42,56,8,96</td><td></td><td>2</td><td></td><td></td><td></td><td></td></tr>
<tr><td>139</td><td>10</td><td>Consider the hash table of size 11. Using quadratic probing, insert the key 126, 75, 37, 56, 29, 154, 10 and 99 into hash table.</td><td></td><td>2</td><td></td><td></td><td></td><td></td></tr>
<tr><td>140</td><td>10</td><td>Consider the hash table of size 10. Using quadratic probing, insert the keys 72, 27, 36, 24, 63, 81, and 101 into hash table. Take c1=1 and c2=3.</td><td></td><td>7</td><td></td><td></td><td></td><td></td></tr>
<tr><td>141</td><td>10</td><td>Consider a hash table of size 11. We want to insert keys 20, 34, 45, 70, 56 in the hash table. Let's insert the keys into hash table using the following double hash functions:
h1(k) = k mod 11 (first hash function)
h2(k) = 8 - (k mod 8) (second hash function)</td><td></td><td>3</td><td></td><td></td><td></td><td></td></tr>
</table>

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
142	10	Consider inserting the keys 76,26,37,59,21,65 into a hash table of size m= 11 using double hashing.consider hash functions are h1(k) = kmod 11 and h2(k)=k mod 9		2				
143	10	Consider a double hashing scheme in which the primary hash function is h1(k)=k mod 23 and the secondary hash function is h2(k)=1+(k mod 19). Assume that the table size is 23. then the address returned by probe 1 in the probe sequence(assume that the probe sequence begins at probe 0) for key value k=90 is____.		3				
144	10	Insert the following keys in an array of size 17 using the modulo division method. Use double hashing to resolve collisions. Take h'(k)=(key%7)+1 as the second hash function: 94,37,29,40,84,88,102,63,67,120,122. What is the value present at location 7 ____		2				